

Name of Student:

Scholar No. :

Indian Institute of Information Technology, Bhopal

Department of IT

End Term Examination, December 2025

Course: B. Tech

Semester: III

Subject: Design and Analysis of Algorithms

Subject Code: IT-3004

Time: 180 Minutes

Max. Marks: 60

Note: Attempt all questions. All questions carry equal marks.

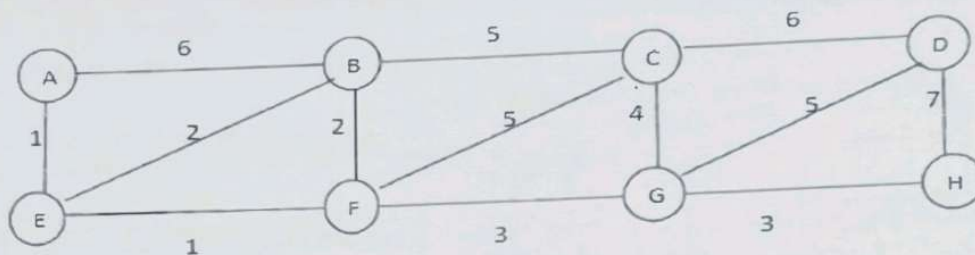
Q. No.	Questions	Marks
1(a).	(i) Can the master method be applied to the recurrence $T(n) = 64T(n/2) - n^2 \log n$? why or why not? Give an asymptotic upper bound for this recurrence. (ii) How are Big O and Big Omega notations graphically represented on the same graph to depict the upper and lower bounds of an algorithm's time complexity? Explain the key features of such a graphical representation.	3+3
1(b).	(i) Suppose you have a computer that requires 1 minute to solve problem instances of size $n = 1,000$. Suppose you buy a new computer that runs 1,000 times faster than the old one. What instance sizes can be run in 1 minute, assuming the following time complexities $T(n)$ for our algorithm? A. $T(n) = n$ B. $T(n) = n^3$ (ii) Using the definitions of O and Ω, show that $6n^2 + 20n \in O(n^3)$ but $6n^2 + 20n \notin \Omega(n^3)$	3+3
2(a).	Given two sorted sub-array $A[p \dots q]$ and $[q + 1, \dots, r]$. Write an algorithm to Merge the two sorted sub-arrays into one sorted array $A[p, \dots, r]$ i.e. MERGE(A, p, q, r). Analyze the running time of the algorithm also.	6
2(b).	(i) Write a divide-and-conquer algorithm for the Towers of Hanoi problem. The Towers of Hanoi problem consists of three pegs and n disks of different sizes. The object is to move the disks that are stacked, in decreasing order of their size on one of the three pegs to a new peg using the third one as a temporary peg. The problem should be solved according to the following rules: (1) when a disk is moved, it must be placed on one of the three pegs; (2) only one disk may be moved at a time, and it must be the top disk on one of the pegs; and (3) a larger disk may never be placed on top of a smaller disk. Show for your algorithm that $S(n) = 2^n - 1$. (Here $S(n)$ denotes the number of steps (moves), given an input of n disks.) (ii) Suppose that, in a divide-and-conquer algorithm, we always divide an instance of size n of a problem into 3 sub instances of size $n/4$, and the dividing and combining steps take a time in $\Theta(n^2)$. Write a recurrence equation for the running time $T(n)$, and solve the equation for $T(n)$.	3+3
3(a).	(i) Formulate the Knapsack problem with greedy method and find the optimal solution for (Object) $n=7$, (Maximum Capacity) $m=15$, Profit $(p_1-p_7)=(10,5,15,7,6,18,3)$, and Weight $(w_1-w_7)=(2,3,5,7,1,4,1)$. (ii) Construct a Huffman tree resulting code word for the following:	3+3

Character	A	B	C	D	-
Probability	0.35	0.1	0.2	0.2	0.15

Encode the words DAD and ADC

3(b).

Consider the following graph



- What is the cost of its minimum spanning tree?
- How many minimum Spanning trees does it have?
- Suppose Kruskal's algorithm is run on this graph. In what order are the edges added to the MST

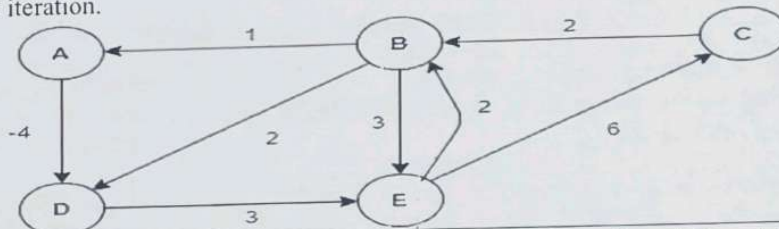
4(a).

Write a MATRIX-CHAIN_ORDER algorithm using top-down approach. Consider the following matrices whose orders are given and perform chain matrix multiplication using dynamic programming.

A	B	C	D
4x5	5x3	3x2	2x7

4(b).

Apply Floyd-warshall algorithm for constructing shortest path. Show the matrix $D^{(k)}$ that results each iteration.



5(a).

Using the distance matrix below, solve the TSP using the Branch and Bound technique. Show all intermediate cost matrix reductions, branching steps, and bounding calculations.

From / To	A	B	C	D
A	∞	20	42	35
B	20	∞	30	34
C	42	30	∞	12
D	35	34	12	∞

Determine:

- The optimal tour (minimum-cost path).
- The lower bound for each branching step.
- The time and space complexity.

5(b).

Discuss 8-queen problem. Write the steps to solve 8-queen problem by backtracking. Write its complexity